[Sandia National Laboratories Aerial Photograph]

Place `cover_photo.jpg` in the same folder
and uncomment the includegraphics line above.

UC DAVIS

College of Engineering

Passive Temperature Compensation Device

SPONSORED BY: DR. BRADLEY BON, SANDIA NATIONAL LABORATORIES

PRELIMINARY MACHINE ARCHITECTURE REPORT

EME-185A/B

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SUBMISSION DEADLINE: 16 MARCH 2026

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Passive Temperature Compensation Device

By the

Noble Aggie Labs

Sponsored by Dr. Bradley Bon, Sandia National Laboratories

Preliminary Machine Architecture Report

EME-185A/B

Professor Jonathan Schofield

TA Xiangpu Wang

Due 16 March 2026

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1 Abstract / Executive Summary

A variety of complex mechanical systems are dependent on electronic components and mechatronic designs requiring sensors, microcontrollers, etc. Sandia National Laboratories (SNL) has tasked our senior design team, Noble Aggie Labs with designing a passive temperature compensation device (PTCD) for the purpose of a robust system that will function without reliance on electricity or humans. The device must respond to ambient air temperature to precisely control the flow of high-pressure gas. The device will adjust the effective orifice diameter according to the Governing linear equation:

$$d_{\text{eff}} = A_1(T - T_{\text{ref}}) + A_0 \tag{1}$$

Be advised, $A_1 = -6.0 \times 10^{-5} \pm 5\%$ in $/^\circ$C, $A_0 = 0.01 \pm 5\%$ in, and $T - \text{ref} = 25\,^\circ$C. The device must operate between $-60\,^\circ$C and $80\,^\circ$C, withstand pressure differentials up to 10 000 psi during choked flow, fit within a $(10\,\text{cm})^3$ mechanical envelope, and be compatible with hydrogen gas.

Over the course of this project, our team curated five actuator concepts and ten valve concepts. All design concepts were then systematic screened and weighted with various scoring matrices (Report 1). We have since converged on a single architecture consisting of three primary subsystems: (1) a **aluminum thermal expansion rod** a singular metallic rod that responds to ambient temperature changes. The aluminum rod will have linear piston displacement as a result of thermal exapnsion via the following equation:

$$\Delta L = L_0\,\Delta T\,\alpha \tag{2}$$

(2) A **compound spur-gear train** comprising of two rack gears and two coaxial pinion gears with a gear ratio of 4:1. This amplifies and transmits the piston's linear motion to the valve; and (3) a **sliding-plate orifice valve** ("pizza cutter" design), in which two flat plates, one fixed and one translating contain a shaped hole with the overlap between the two holes defining the effective flow orifice. As the actuator extends with increasing temperature, the moving plate translates downwards. This reduces the overlap area and shriking d_{eff} per the requirements.

2 System Architecture Selected

2.1 Selection Criteria and Design Justification

[TODO: Include volume specification and volume of subsystems]

[TODO: State the single concept being carried forward. For example: "After evaluating five actuator concepts and ten valve concepts through systematic screening and weighted scoring matrices (see Appendix B), our team selected the [Actuator Type] coupled with a [Valve Type] as our final system architecture."]

[TODO: Discuss the selection criteria used:

- **Temperature range compatibility ($-60\,^{\circ}\mathrm{C}$ to $80\,^{\circ}\mathrm{C}$)**
- **Mechanical envelope ($10\,\mathrm{cm}$ cube)**
- **Hydrogen compatibility of wetted materials**
- **Ability to withstand pressure differentials up to $10\,000\,\mathrm{psid}$**
- **Linear or controllable relationship between temperature and flow restriction**
- **Simplicity and reliability (passive operation, no electronics)**

]

[TODO: Explain why each subsystem solution was chosen. Emphasize STRENGTHS, not weaknesses of alternatives.]

2.2 Complete System Overview

[TODO: Provide a 1–2 paragraph overview of how the entire system works at a high level. Walk the reader through the flow of energy/information: temperature change → actuator displacement → valve position change → flow restriction adjustment.]

[Full System Assembly — Isometric View]

Include arrows pointing to each subsystem.
Label: Actuator, Valve, Housing, Inlet, Outlet.

Figure 1: *Isometric view of the complete TCD assembly. Arrows indicate the actuator subsystem (left) and valve subsystem (right). Flow enters from the inlet port and exits through the outlet port.*

[TODO: Reference the figure: "As shown in Figure 1, the system consists of. . ."]

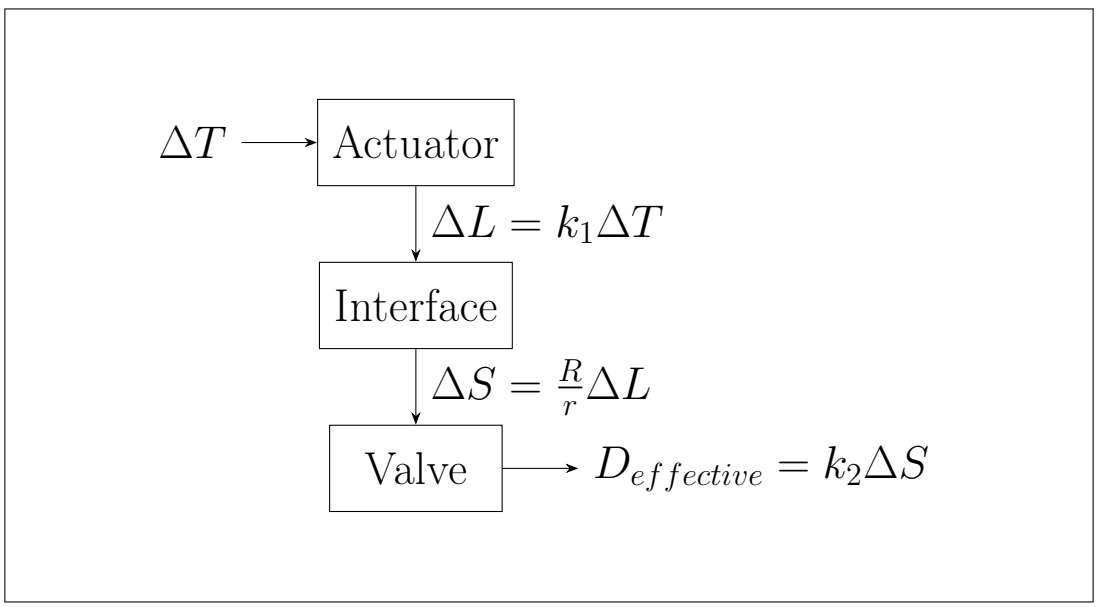

Figure 2: *Controls diagram of the TCD. Outlines the overall flow of inputs and outputs*

2.3 Subsystem Overviews

2.3.1 Thermal Actuator

[TODO: Describe the actuator subsystem. Include:

- **Operating principle (how temperature change creates displacement)**
- **Key components and their functions**
- **Why this actuator type was selected (reference scoring matrix)**
- **Expected displacement range over the temperature range**

]

[Actuator Sub-Assembly Drawing]

Exploded view with numbered callouts.
Show how displacement is generated.

Figure 3: *Actuator sub-assembly. The numbered components are: (1)* ***[TODO: component]****, (2)* ***[TODO: component]****, (3)* ***[TODO: component]****.*

2.3.2 Valve

[TODO: Describe the valve subsystem. Include:

- **Valve type and operating principle**
- **How actuator displacement translates to flow area change**
- **Key components**
- **Expected flow coefficient (Cv) range**

]

[Valve Sub-Assembly Drawing]

Show open and closed positions.
Use dashed lines for alternate positions.

Figure 4: *Valve sub-assembly showing the flow restriction mechanism.*

2.3.3 Housing and Fluid Interface

[TODO: Describe the housing that contains everything. Include:

- **Material selection (hydrogen compatibility, steel/aluminum)**
- **Sealing strategy (O-rings, gaskets, welding)**
- **How the device connects to the existing piping system**
- **Thermal conduction path from ambient air to the actuator**

]

3 Discussion

3.1 Valve

3.1.1 Bellows Description and Calculations

The welded metal bellows transmits an axial stroke δ from an external actuator into a high-pressure hydrogen valve while maintaining zero leakage. The bellows must withstand internal pressure P, provide the required displacement without plastic deformation, and minimize actuator force demand.

Design constraints:

- Withstand maximum internal pressure

- Operate within allowable compression limits
- Maintain acceptable spring force relative to actuator capability
- Minimize pressure-induced axial force via geometry

Design parameters:

- $P = 10{,}000$ psi (maximum internal pressure)
- $\delta = 0.01053$ in (required stroke)

INSERT BELLOWS IMAGE HERE

Figure XXX, above, illustrates the valve assembly with the bellows highlighted. Two bellows are used in the design to negate the pressure-induced force, allowing the actuator to focus solely on overcoming spring force and friction. They are welded to the valve body and the valve stem, ensuring a leak-tight seal while allowing for the necessary axial movement. Both bellows are designed to be identical, so the calculations below apply to each bellows independently. The following equations consider important design parameters and constraints for selecting the appropriate bellows geometry and material.

Effective area:

$$A = \pi \left(\frac{D_o + D_i}{4} \right)^2$$

Where D_o is the outside convolution diameter and D_i is the inside convolution diameter of the bellows. The effective area determines the axial force generated by internal pressure. Therefore, minimizing A reduces actuator force demand.

Pressure-induced force:

$$F_P = PA$$

Let K be the spring rate of the bellows, which depends on material properties, geometry, and convolution count N. The spring force is proportional to deflection δ.

Spring force:

$$F_s = K\delta$$

Total force required for actuation:

$$F_{total} = PA - PA + 2K\delta + F_{friction} \pm mg$$

It should be noted that δ is very small, so the spring force contribution is minimal compared to the pressure force. The two identical and opposing bellows cancel out the pressure-induced force. $F_{friction}$ accounts for any additional forces due to friction in the system, which should be minimized through design and material selection. The term mg accounts for gravitational forces acting on the valve stem.

Let N be the number of convolutions, D_{max} be the maximum axial displacement of the bellows, L be the convolution free length, and D_c be the axial displacement per convolution. The total deflection must be less than the maximum allowed to prevent plastic deformation.

$$\delta < D_{max}$$

Displacement per convolution:

$$D_c = \frac{D_{max}}{N}$$

Increasing N lowers spring rate and increases stroke capacity. Decreasing N increases stiffness and squirm resistance. In this application, δ is small, so the spring rate is chosen to be low enough to allow for the required stroke while maintaining structural integrity.

When welded metal bellows are subjected to internal pressure, the pressure acts over an effective cross-sectional area and generates a net axial force. This pressure-induced force is independent of stroke and depends solely on system pressure and bellows geometry. In high-pressure hydrogen applications, this force typically dominates the actuator load requirement and therefore becomes a primary design driver.

The pressure required to compress the bellows any distance within its elastic range is given by:

$$P = \frac{K\delta}{A}$$

The maximum operating pressure of the bellows is considered the proof pressure, which is the pressure at which the bellows can be compressed to the required stroke without yielding. Critical squirm pressure is the pressure at which the bellows becomes unstable and experiences lateral deformation. This occurs when the bellows are unrestrained from sideways movement and the internal pressure exceeds the bellows' ability to maintain its shape.

Burst Pressure is the maximum pressure the bellows can withstand while restrained from lateral movement. For this application, the bellows will be restrained from lateral movement

by the valve body, so critical squirm pressure is not a primary concern. Therefore, the design is governed by the following relationship:

$$P_{operating} < P_{burst}$$

Where P_{burst} is the maximum pressure the bellows can withstand without failure.

The equations above govern the selection process for the bellows parameters and define the acceptable diameter, wall thickness, convolution count, and material. The parameters of the selected bellows are as follows:

- Do: in (outside convolution diameter)
- Di:in (inside convolution diameter)
- t:in (wall thickness)
- L:in (convolution free length)
- Pburst:psi (burst pressure)
- N:(number of convolutions)
- Material:(chosen for its strength and compatibility with hydrogen)

Justification for the selected parameters is based on the formulas above which yield the required stroke, spring force, and pressure resistance. The selected bellows meet the design constraints while minimizing actuator force demand and ensuring reliable operation under the specified conditions.

3.1.2 Design Requirement Updates

[TODO: Discuss how requirements changed since Report 1. For example:

- **Did discussions with Dr. Bradley Bon reveal new constraints?**
- **Did preliminary calculations show any requirements were infeasible or needed adjustment?**
- **Were any requirements added, removed, or modified?**

Reference the updated requirements table in Appendix A.]

3.1.3 Requirements Rolldown to Subsystems

[TODO: Explain how system-level requirements were decomposed into subsystem-level specifications. This is critical for the grading rubric.]

Table 1: *Example requirements rolldown from system to subsystem level.*

System Req.	Spec	Subsystem Rolldown	Sub-Spec
Temperature range	$-60\,^{\circ}\mathrm{C}$ to $80\,^{\circ}\mathrm{C}$	**[TODO: Actuator must produce displacement over this full range]**	**[TODO: spec]**
Mechanical envelope	$\leq 10\,\mathrm{cm}^3$	**[TODO: Actuator $\leq$ X cm, Valve $\leq$ Y cm, Housing $\leq$ Z cm]**	**[TODO: spec]**
Pressure differential	100–10 000 psid	**[TODO: Valve must not deform or leak under max pressure]**	**[TODO: spec]**
H_2 compatibility	All wetted parts	**[TODO: Specify material for each wetted component]**	**[TODO: spec]**

3.1.4 Subsystem Interactions and Technical Detail

[TODO: This is where you go DEEP into the technical details. Unlike Section 2 which was a high-level overview, this section provides engineering proof that the design works.]

[TODO: Describe how subsystems interface with each other:

- **How does the actuator physically connect to the valve?**
- **What is the mechanical coupling (direct contact, linkage, etc.)?**
- **How does thermal energy flow through the system?**
- **What are the potential failure modes at interfaces?**

]

3.1.5 Design Verification Calculations

[TODO: Introduce the purpose of these calculations: to demonstrate analytically that the design can meet its requirements.]

3.1.6 Thermal Expansion and Actuator Displacement

[TODO: Show that the actuator produces sufficient displacement over the temperature range to fully open/close the valve.]

[TODO: Replace this example with your actual thermal expansion calculation.]

The linear thermal expansion of a rod is given by:

$$\Delta L = \alpha \cdot L_0 \cdot \Delta T \tag{3}$$

where:

- ΔL is the change in length [mm],
- α is the coefficient of thermal expansion [$^\circ\mathrm{C}^{-1}$],
- L_0 is the original length [mm],
- ΔT is the temperature change [°C].

  Assumptions:

  1. **[TODO: Material properties are constant over the temperature range.]**
  2. **[TODO: Thermal equilibrium is reached before the test window.]**
  3. **[TODO: List all assumptions.]**

[TODO: Substitute your values into Equation 3 and show the result. Compare the result to the requirement.]

3.1.7 Force Analysis Under Pressure Loading

[TODO: Show that the actuator can generate enough force to move the valve against the pressure differential. Consider worst case: 10 000 psid.]

Paramter values:

- $E =$

- $A =$
- $\alpha =$
- $\Delta T =$

$$F = EA\alpha\Delta T \tag{4}$$

[TODO: Insert your actual force/pressure calculations here.]

3.1.8 Performance Relationship

[TODO: Demonstrate that the system achieves the required flow characteristic (e.g., the relationship between temperature and effective flow area or discharge coefficient). Reference Performance_Relationship.xlsx if applicable.]

3.1.9 Additional Analyses

[TODO: Add any other calculations that prove feasibility:

- **Heat transfer analysis (thermal response time)**
- **Stress/strain analysis of critical components**
- **Material compatibility verification**
- **Tolerance stack-up analysis**
- **FEA results (if performed)**

]

4 Preliminary Bill of Materials

[TODO: Write a brief introductory paragraph explaining the BOM structure and the total estimated project cost.]

Table 2: *Preliminary Bill of Materials and Cost Estimate*

Item	Part Name	Description / Supplier	Qty	Material	P/
Actuator Subsystem					
1	[TODO: Part name]	[TODO: Description / McMaster #]	[TODO: 1]	[TODO: Steel]	P
2	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: Al 6061]	F
Valve Subsystem					
3	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: SS 316]	P
4	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: Brass]	F
Housing & Interface					
5	[TODO: Part name]	[TODO: Description]	[TODO: 1]	[TODO: SS 316]	F
6	[TODO: Part name]	[TODO: Description]	[TODO: 2]	[TODO: Viton]	P
Fasteners & Miscellaneous					
7	[TODO: Part name]	[TODO: Description]	[TODO: 10]	[TODO: Steel]	P
Special Tooling					
8	[TODO: Tooling item]	[TODO: Description]	[TODO: 1]	—	P
				Estimated Tota	

Key: P = Purchased, F = Fabricated

[TODO: Add any notes about budget constraints, contingency funds, or items awaiting price quotes.]

5 Project Management Information

5.1 Updated Task List

[TODO: Provide a detailed task list. Organize by phase or subsystem. Must include:

- **Technical tasks (design, fabrication, testing)**
- **Deliverables for class (reports, presentations)**
- **Team and stakeholder meetings**
- **Procurement / ordering deadlines**

]

Table 3: *Updated task list for Winter and Spring quarters.*

#	Task Description	Owner(s)	Deadline
1	Finalize engineering drawings	**[TODO: Name(s)]**	**[TODO: Date]**
2	Order raw materials	**[TODO: Name(s)]**	**[TODO: Date]**
3	Machine actuator housing	**[TODO: Name(s)]**	**[TODO: Date]**
4	Assemble prototype	**[TODO: Name(s)]**	**[TODO: Date]**
5	Conduct thermal testing	**[TODO: Name(s)]**	**[TODO: Date]**
6	Conduct pressure testing	**[TODO: Name(s)]**	**[TODO: Date]**
7	**[TODO: Add more tasks]**		

5.2 Gantt Chart

[TODO: Insert your Gantt chart. This is typically created in Excel, MS Project, or a similar tool and exported as an image.]

[Gantt Chart — Two Quarters]

Export from Excel/MS Project as PNG or PDF.
Must span Winter 2026 through Spring 2026.
Show current progress with a vertical "today" line.

Figure 5: *Updated Gantt chart spanning Winter and Spring quarters. The vertical dashed line indicates the current date.*

[TODO: Write a paragraph interpreting the Gantt chart. Highlight key milestones and any tasks that are ahead of or behind schedule.]

5.3 Critical Path Method (CPM) Analysis

[TODO: Insert your CPM network diagram. This must be in GRAPHICAL format.]

[CPM Network Diagram]

Show tasks as nodes, dependencies as arrows.
Highlight the critical path in red or bold.
Include earliest start/finish and latest start/finish times.

Figure 6: *Critical path method (CPM) network diagram. The critical path is highlighted in red.*

[TODO: Write a paragraph describing:

- **What the critical path is (list the tasks)**
- **Total project duration along the critical path**
- **Which tasks have float (slack time)**
- **What the analysis tells you about risk and scheduling**
- **Any mitigation strategies for critical-path delays**

]

6 Cited References

References

[1] B. Bon, "Passive Temperature Compensation Device — Project Prompt," Sandia National Laboratories, SAND2025-15446O, 2025.

[2] **[TODO: Add reference: textbook, paper, or technical standard]**

[3] **[TODO: Add reference: material property data source]**

[4] **[TODO: Add reference: relevant prior work or patent]**

Appendices

A Updated Design Requirements

[TODO: Include the final, updated requirements table here. This should reflect any changes discussed in Section 3.1.2. Organize by category (system-level, actuator, valve, housing, etc.). Include the subsystem-level rolldown specifications.]

B Concept Screening and Scoring Matrices

[TODO: Include the screening matrix and weighted scoring matrix from your concept selection process. These were referenced in Section 2.1.]

C Engineering Drawing Package

[TODO: Include complete assembly, sub-assembly, and part drawings. Part drawings must include dimensions and tolerances for all parts that will be MANUFACTURED (not purchased). Follow proper engineering drawing practices.

Organize as:

1. **Full system assembly drawing**
2. **Actuator sub-assembly drawing**
3. **Valve sub-assembly drawing**
4. **Housing sub-assembly drawing**
5. **Individual part drawings (manufactured parts only)**

Each drawing should be on its own page for clarity. Label as Figure A.1, A.2, etc.]

D Product Data Sheets and Price Quotes

[TODO: Include manufacturer spec sheets for all purchased components. Include price quotes or screenshots of supplier pricing. This supports the Bill of Materials in Section 4.]

E Detailed Calculations

[TODO: If your calculations in Section 3.1.5 were summarized for brevity, include the full derivations and spreadsheet screenshots here. Reference Performance_Relationship.xlsx and Thermal_Expansion.xlsx data as appropriate.]